Yarroju Rithvik

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Education

M.Tech, Aerodynamics and Flight Mechanics Aug 2024 – Jul 2026
Indian Institute of Space Science and Technology, Thiruvananthapuram *CGPA: 8.20*

- **Major Thesis:** Machine Learning for Aerospace Applications.
- **Relevant coursework:** Multidisciplinary Design Optimization; Space Flight Mechanics; Aerodynamics; Flight Mechanics; Propulsion; Compressible Flow.

M.Tech, Applied Artificial Intelligence *(Online)* Aug 2024 – Jul 2026
Visvesvaraya National Institute of Technology (VNIT), Nagpur *CGPA: 7.45*

- **Major Thesis:** Kolmogorov-Arnold Networks for Interpretable Reinforcement Learning and Remote Sensing.
- **Relevant coursework:** Computer Vision; Machine Learning Algorithms; SQL; Big Data; MLOps; Neural Networks.

B.Tech, Chemical Engineering Aug 2016 – Jul 2020
Indian Institute of Petroleum and Energy (IIPE), Visakhapatnam *CGPA: 8.11*

- **Relevant coursework:** Data Analytics and AI; Probability and Statistics; Programming and Data Structures; Information Technology.

Projects

Aerostat AI Stabilization | *Python, Sensor Fusion, Motion Prediction, Control Systems*

- Developed a simulation-based AI pipeline for stabilizing a tethered aerostat under wind disturbance scenarios.
- Built sensor-fusion and motion-prediction components for estimating aerostat motion and disturbance-aware system behaviour.
- Designed a modular control-oriented workflow for testing stabilization logic under changing wind conditions.

Physics-Informed KANs for Longitudinal Drag Model Identification | *Python, PyTorch, KAN, Physics-Informed ML*

- Developed physics-informed KAN models for Hansa aircraft drag model identification using the longitudinal speed-equation residual.
- Learned an interpretable drag-coefficient relation $C_D(\alpha, \delta_e)$ from trajectory data and compared it with classical aerodynamic parameter-estimation baselines.
- Applied KAN sparsification, pruning, grid refinement, and symbolic replacement to obtain a compact and interpretable aerodynamic model.

Unsupervised Remote Sensing Change Detection using KANs | *Python, PyTorch, Remote Sensing, Computer Vision*

- Adapted a published Metric-CD-style unsupervised change-detection pipeline and replaced Polynomial Color Correction with a KAN-based nonlinear harmonization module for bi-temporal satellite image pairs.
- Designed change-detection evidence channels using harmonized RGB differences, signed differences, raw differences, and gradient-based differences.
- Integrated KAN-based harmonization with a Gated Enhanced ResNet pipeline for unsupervised changed-pixel detection on remote sensing image pairs.

Technical Skills

Languages: Python, SQL, C/C++, MATLAB
ML & Data Science: NumPy, Pandas, scikit-learn, CatBoost, Regression, Classification, Clustering, Model Evaluation
Deep Learning & Computer Vision: PyTorch, TensorFlow/Keras, OpenCV, CNNs
MLOps & Deployment: Git, Docker, FastAPI, Streamlit, MLflow, Experiment Tracking, End-to-End ML Workflows
Research & Engineering AI: Kolmogorov-Arnold Networks, Physics-Informed ML, Reinforcement Learning, Optimization

Experience & Achievements

Data Science & Machine Learning Intern Feb 2026 – May 2026
Finlatics – Remote

- Built Python-based ML projects across banking classification, social-media clustering, and sales prediction; modeled 45,216 banking records using Logistic Regression/Random Forest and clustered 7,050 engagement records using K-Means.

Achievements & Test Scores: JEE (2016) — Cleared both Mains and Advanced | GRE — 326 | IELTS — 7.5